

Education

Virginia Tech University

August 2024 - Present

Major: Computational Modeling and Data Analytics

Specialization: Cryptography

Minors: Computer Science and Mathematics

Sample Courses: R Programming (Harvard Certificate), Leadership (MIT Certificate), Data Science (Harvard, Non-Certification), Data Structures (HS), Mobile Programming (HS), Advanced Statistics & Probability (HS)

Projects

Data Structure Projects | (Coded in Java)

- **Dijkstra Algorithm:** To replicate GPS functionalities, I designed a Dijkstra-based system to calculate shortest paths and distances between major cities, mimicking a real-life navigation solution for route planning.
- **Complex Dynamics Visualization Program:** To enhance the visualization of complex mathematical functions, I created an interactive GUI to graph Julia and Mandelbrot sets to make complex functions more accessible for educational purposes.
- **Binary Search Trees:** Faced with the challenge of arranging large, disorganized datasets, I developed a solution using a *Binary Search Tree* with multiple traversal methods creating an organization system with swift data retrieval speeds.
- **Superlist Project:** Directed to integrate multiple data structures into a single, cohesive list. I created a "Superlist" utilizing *List Nodes* to inherit the functionalities of Stacks, Arrays/Array Lists, and Queues, simplifying multiple data structures within a singular *superclass*.

Financial / Technical Projects | (Coded in Pine Script and Python)

- **Dual Polynomial Projection:** To predict future price movements, developed a model that projects multinomial curves beyond observed data points that provided valuable insights into potential market trends.
- **The Tiny Oracle:** Needing to assess price movements, created an analytical tool that evaluates multiple financial indicators and delivers a "Buy, Sell, or Neutral" rating, providing actionable insights on market trends.
- **Refined Time-Series Analysis:** Having to forecast potential trend shifts, implemented a predictive tool using Linear regression and Time-Series Analysis with adjustable periods to enhance trend predictions.

Experience

FOUNDER | [ALGO SYNTHESIS](#) | SEPTEMBER 2022 - PRESENT

Build trading strategies to gain hands-on experience in programming, trading, and business development.

- Developed over 7 full-cycle trading indicators and strategies, optimizing algorithms and improving trading efficiency, and achieving various success and profit ratios.
- Mentored 5 classmates and significantly improved their interest, engagement, and future progression into in programming and trading.

BARISTA & TEAM TRAINER | STARBUCKS BISTRO | JUNE 2023 – JULY 2024

- Created positive work environment. Helped resolve conflicts. Assisted customers. Opening/Closing responsibilities. Cash register duties. Prepared drinks.

Technical Skills

Programming Languages: Java, Python, R, Pine Script, JavaScript, TypeScript

Tools & Technologies: GitHub, Bash, Stack Overflow, IntelliJ, Visual Studio, TradingView, Android Studio.

Administrative: Office 365 (Excel, Word, Outlook, PowerPoint, Teams), Google G Suite, Zoom.

Interests: Coding, Quant, Financial Markets, Reading, Weightlifting, Volleyball, Tennis, Rowing, Hiking.